Database Assignment 1 solutions

1. create table Authors (author_id INT Primary Key, name VARCHAR(100), country VARCHAR(50));
   CREATE TABLE Books (
   -> book_id INT PRIMARY KEY,
   -> title VARCHAR(150),
   -> author_id INT,
   -> FOREIGN KEY (author_id) REFERENCES Authors(author_id),
   -> published_year YEAR,
   -> available_copies INT
   ->);
   create table Members (member_id INT PRIMARY KEY, name VARCHAR(100), email VARCHAR(100) UNIQUE, phone VARCHAR(15));
CREATE TABLE Borrow_Records (
   -> record_id INT PRIMARY KEY,
   -> member_id INT,
   -> book_id INT,
   -> borrow_date DATE,
   -> return_date DATE,
   -> FOREIGN KEY (member_id) REFERENCES Members(member_id),
   -> FOREIGN KEY (book_id) REFERENCES Books(book_id)
   ->);

2. create table Authors (author_id INT Primary Key, name VARCHAR(100), country VARCHAR(50));

3. CREATE TABLE Books (
   -> book_id INT PRIMARY KEY,
   -> title VARCHAR(150),
   -> author_id INT,
   -> FOREIGN KEY (author_id) REFERENCES Authors(author_id),
   -> published_year YEAR,
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   ->);

4. create table Members (member_id INT PRIMARY KEY, name VARCHAR(100), email VARCHAR(100) UNIQUE, phone VARCHAR(15));

5. CREATE TABLE Borrow_Records (

```
  ->    record_id INT PRIMARY KEY,
  ->    member_id INT,
  ->    book_id INT,
  ->    borrow_date DATE,
  ->    return_date DATE,
  ->    FOREIGN KEY (member_id) REFERENCES Members(member_id),
  ->    FOREIGN KEY (book_id) REFERENCES Books(book_id)
  -> );
```

6.
```
alter table books
  -> add column genre VARCHAR(50);
```

7.
```
drop table borrow_records;
```

8.
```
insert into authors (author_id, name, country) values (1, 'Rutuj', 'India');
  insert into authors (author_id, name, country) values (2, 'Sameer', 'UK');
  insert into authors (author_id, name, country) values (3, 'Rajiv', 'USA');
```

9.
```
INSERT INTO Books
  -> (book_id, title, author_id, published_year, available_copies)
  -> VALUES
  -> (1, 'Wings of Fire', 1, 1999, 5),
  -> (2, 'Jungle Book', 2, 1894, 10),
  -> (3, 'The Alchemist', 2, 1988, 3),
  -> (4, 'Harry Potter', 3, 1997, 4),
  -> (5, 'The Hobbit', 4, 1937, 2);
```

10.
```
insert into Members (member_id, name, email, phone) values
  -> (1, 'Rutuj', 'rutuj@gmail.com', '1111111111'),
  -> (2, 'Om', 'om@gmail.com', '2222222222'),
  -> (3, 'Prakash', 'prakash@gmail.com', '333333333333');
```

11.
```
insert into Borrow_Records (record_id, member_id, book_id, borrow_date,
return_date) values (1, 1, 1, '2026-10-01', '2026-10-10'), (2, 2, 2, '2026-10-03', '2026-10-
10'), (3, 3, 3, '2026-10-05', NULL), (4, 1, 4, '2026-10-07', NULL);
```

12.
```
select * from books where published_year > 2015;
```

13.
```
 CREATE TABLE department (
  dept_id INT PRIMARY KEY,
  dept_name VARCHAR(50)
```

```
);
CREATE TABLE student (
  student_id INT PRIMARY KEY,
  name VARCHAR(50),
  dept_id INT,
  FOREIGN KEY (dept_id) REFERENCES department(dept_id)
);
```

14. select * from Borrow_Records where book_id = 6;

15.
```
update Books
  -> set available_copies = available_copies - 1
  -> where book_id in (
  -> select book_id
  -> from Borrow_Records
  -> );
```

16.
```
SELECT *
  -> FROM borrow_records
  -> WHERE member_id = 3;
DELETE FROM borrow_records
  -> WHERE member_id = 3;
```

17.
```
UPDATE books
  -> SET title = 'Words of Fire'
  -> WHERE book_id = 1;
```

18.
```
UPDATE books
  -> SET author_name = 'a'
  -> WHERE author_id = 1;
UPDATE books
  -> SET author_name = 'b'
  -> WHERE author_id = 2;
UPDATE books
  -> SET author_name = 'c'
  -> WHERE author_id = 3;
UPDATE books
  -> SET author_name = 'd'
  -> WHERE author_id = 4;
UPDATE books
  -> SET author_name = 'e'
  -> WHERE author_id = 5;
```

```
UPDATE books
   -> SET author_name = 'f'
   -> WHERE author_id = 6;

19. DELETE FROM borrow_records
   -> WHERE book_id IN (
   ->    SELECT book_id
   ->    FROM books
   ->    WHERE published_year < 2000
   -> );

20. SELECT *
   -> FROM books
   -> WHERE book_id NOT IN (
   ->    SELECT book_id
   ->    FROM borrow_records
   -> );
```